Stage 5: Clinical Information Extraction with BioBERT

Theory notes, model choice and interview questions
Project: Doctor–Patient Conversation Audio -> SOAP Notes

What this stage does: runs fine-tuned BioBERT token-classification (NER) models on every DOCTOR / PATIENT utterance and extracts clinical entities with their exact character positions, confidence and the speaker who said them. Labels are reported exactly as the models predict them: **DISEASE** and **CHEMICAL**. DURATION and SEVERITY are not supported by any BioBERT checkpoint and are reported as such.

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1. What Named Entity Recognition (NER) is

NER finds spans of text that refer to things of a given type and labels them:

```
I have had  fever  for three days and a severe  headache .
          [DISEASE]                            [DISEASE]
```

- It is a **token classification** task: every token receives a label; consecutive labelled tokens form an entity.
- Clinical NER is the first step of turning free conversation into structured data (symptoms, drugs, conditions) that a SOAP note can be built from.
- NER only **finds** mentions. It does not decide if a symptom is present, absent ('no fever') or belongs to someone else - those are separate tasks (negation / assertion detection).

2. BERT in one page

- **BERT** (Bidirectional Encoder Representations from Transformers, Google 2018) is a stack of **Transformer encoder** layers. BERT-base: **12 layers, hidden size 768, 12 attention heads, about 110 M parameters**.
- **Bidirectional**: each token's vector is computed by attending to all other tokens on both sides, so the representation of 'cold' differs in 'I have a cold' vs 'the room is cold'. These are **contextual embeddings**.
- **Pre-training** (self-supervised, no labels): **Masked Language Modelling** - hide 15% of tokens and predict them; and **Next Sentence Prediction**. This teaches grammar and meaning from raw text.
- **Fine-tuning**: add a small task-specific layer on top and train the whole network on labelled data for a task (NER, classification, QA).
- Special tokens: **[CLS]** at the start, **[SEP]** at the end. Text is split into **WordPiece** sub-words ('migraine' -> 'mi', '##gra', '##ine').

3. What makes BioBERT different

	BERT-base (cased)	BioBERT (DMIS Lab, Korea Univ., 2019)
Starting point	random	initialised from BERT-base cased
Pre-training text	Wikipedia + BooksCorpus (3.3 B words)	+ PubMed abstracts (4.5 B words) + PMC full-text articles (13.5 B words)
Vocabulary	BERT WordPiece	same WordPiece vocabulary as BERT (kept for compatibility)
Architecture	12 layers, 768, 110 M params	identical
Strength	general English	biomedical terminology and phrasing

- BioBERT is a **domain-adapted language model**: same architecture, continued pre-training on biomedical text. It outperformed BERT on biomedical NER, relation extraction and question answering.
- **Base BioBERT has no NER ability by itself.** It only produces contextual vectors. To extract entities it must be fine-tuned with a token-classification head on a labelled NER dataset - which is what our two checkpoints are.

4. From BioBERT to an NER model: fine-tuning

```
BertForTokenClassification
  bert        = BioBERT encoder (12 Transformer layers)   -> (tokens, 768)
  dropout     = 0.1 during training (off at inference)
  classifier  = Linear(768 -> num_labels)                 -> (tokens, 3)  B-DIS, I-DIS, O

Training: cross-entropy loss between predicted and gold label for each token,
          back-propagated through the classifier AND all BioBERT layers.
```

Our checkpoints (by alvaroalon2, Apache-2.0):

Checkpoint	Fine-tuned on	Labels
alvaroalon2/biobert_diseases_ner	BC5CDR-disease (1,500 PubMed abstracts) + NCBI-disease (793 abstracts)	B-DISEASE, I-DISEASE, '0' (outside)
alvaroalon2/biobert_chemical_ner	BC5CDR-chemical + BC4CHEMD (10,000 PubMed abstracts)	B-CHEMICAL, I-CHEMICAL, O

5. BIO tagging and sub-word tokens

```
Text   : Have you had any nausea or vomiting ?
Tokens : [CLS] Have you had any nausea or vomit ##ing ? [SEP]
Labels :   -    O    O   O   O  B-DIS  O B-DIS I-DIS O   -
Entities: 'nausea' (DISEASE), 'vomiting' (DISEASE)
```

- **B-X** = beginning of an entity of type X, **I-X** = inside (continuation), **O** = outside. B is needed to separate two adjacent entities of the same type.
- **Sub-word rule**: a word takes the label of its **first sub-token** - the standard convention used when BERT NER models are trained. 'vomit ##ing' becomes the word 'vomiting'.
- **Character offsets** come from the fast tokenizer's offset_mapping, so each entity points exactly into the original sentence (we verified text[start:end] == entity text for every entity).
- **Confidence** = softmax probability of the predicted label, averaged over the entity's words. It is the model's real output, not an invented score - but neural networks are often over-confident (we see 1.00 a lot), so it is not a calibrated probability of being correct.
- Quirk: the disease checkpoint names its outside label '0' (zero) instead of 'O'; the code accepts both.

6. Our pipeline, step by step (clinical_extraction.py)

Step	Code	Output
Tokenizer	tokenizer(text, return_offsets_mapping=True)	WordPiece ids, char offsets, word ids
Encoder	model.bert(**enc).last_hidden_state	(1, tokens, 768) contextual vectors
Classifier	model.classifier(model.dropout(hidden))	(1, tokens, 3) logits
Softmax	logits.softmax(-1)	probability per label per token
Word labels	first sub-token of each word	label + probability per word
BIO decoding	B starts, I continues, O ends	entities with start/end/confidence

Step	Code	Output
Speaker context	attach role + diarization label + time	conversation[] in JSON

We call the encoder and classifier separately (instead of the one-line transformers pipeline) to make each stage visible; we verified the result is identical to calling the full model. Both models run on every utterance and their entities are merged and sorted by position. Input: outputs/<name>_roles.json or a text file of 'ROLE: text' lines. Output: outputs/<name>_clinical_entities.json. --raw prints every token's label and probability.

7. Model selection: what exists on Hugging Face

Checkpoint	Labels	Decision
siddharthtumre/biobert-finetuned-ner (most downloaded)	DNA, RNA, protein, cell_line, cell_type	rejected: molecular biology, not clinical
alvaroalon2/biobert_diseases_ner	DISEASE	**chosen**: named datasets, Apache-2.0
alvaroalon2/biobert_chemical_ner	CHEMICAL	**chosen**: drugs, Apache-2.0
ghadeermobasher/BC5CDR-Chemical-Disease...	Disease, Chemical	rejected: no model card or license
pmaitra/en_biobert_ner_symptom	symptom	rejected: spaCy wheel, unverifiable 99.95% F1
MACCROBAT models (many symptom/duration labels)	Sign_symptom, Duration, Severity ...	rejected: DistilBERT / DeBERTa, not BioBERT
dmis-lab/biobert-base-cased-v1.2	none (no NER head)	base model only - needs fine-tuning

8. Label mismatch: what we can and cannot extract

Project category	Supported?	How it appears
SYMPTOM	partly	under DISEASE - BC5CDR follows MeSH, which annotates signs and symptoms as diseases; symptom vs disease is not distinguished
DISEASE / CONDITION	yes	DISEASE
MEDICATION	mostly	CHEMICAL - includes drugs but also any chemical
DURATION	no	listed under unsupported_labels in the JSON
SEVERITY	no	listed under unsupported_labels in the JSON

> We deliberately **do not rename** DISEASE to SYMPTOM or CHEMICAL to MEDICATION: that would claim a capability the models do not have. Any later mapping must be an explicit, documented post-processing step.

9. Results and observations

Utterance	Entities found
PATIENT: I have had fever for three days and a severe headache.	fever (DISEASE), headache (DISEASE)
DOCTOR: Do you have a cough?	cough (DISEASE)
PATIENT: I took paracetamol.	paracetamol (CHEMICAL)
PATIENT: My head has been hurting since yesterday, and I feel dizzy.	dizzy only - 'head ... hurting' missed
DOCTOR: Have you had any nausea or vomiting? PATIENT: Yes.	nausea, vomiting - but said by the DOCTOR
DOCTOR: It looks like a migraine. I will prescribe something for the pain.	migraine, pain (DISEASE)

- **Strong on medical nouns** (fever, headache, migraine, paracetamol): confidence about 1.00.
- **Weak on lay descriptions** ('my head has been hurting'): the models learned from scientific abstracts, not conversation - **domain shift**.
- **Speaker attribution is not symptom ownership**: the doctor said 'nausea' in a question; the patient confirmed with 'Yes.'. The SOAP stage must read question-answer pairs, not just entity speakers.
- 'Do you have a cough?' (doctor) and 'migraine' (doctor's assessment) show the same issue: an entity's speaker tells you its *role in the dialogue*, not whether the patient has it.

10. Evaluation of NER

- **Entity-level precision** = correct entities found / all entities found; **recall** = correct found / all true entities; **F1** = harmonic mean. Computed with the seqeval library.
- **Strict match**: same span and type. **Partial / lenient**: overlapping span and same type - useful when 'severe headache' vs 'headache' boundaries differ.
- Published BioBERT results (BC5CDR) are around 85-93 F1 on *abstracts*; on conversations expect lower. A proper evaluation needs a small hand-annotated set of our own consultation transcripts.

11. Limitations

- Domain shift: literature-trained models miss conversational phrasing.
- No negation / assertion: 'no fever' still yields fever. Needs a NegEx-style rule or an assertion model.
- No DURATION / SEVERITY / dosage / frequency labels.
- Symptom vs disease not separated; CHEMICAL is broader than medication.
- No entity linking: 'tummy ache' and 'abdominal pain' are not mapped to the same concept (UMLS / SNOMED CT). DMIS Lab's BioSyn models do this linking for BioBERT.
- Over-confident probabilities; errors from ASR (misheard drug names) pass straight through.
- 512-token limit per input (we process one utterance at a time, so not an issue here).

12. How to get DURATION / SEVERITY (next options)

Option	What	Pros	Cons
A. Rules	regex for 'for three days', 'since yesterday', 'mild/severe'; tagged source='rule'	fast, transparent	not BioBERT; brittle
B. Fine-tune BioBERT	train dmis-lab/biobert-base-cased-v1.2 on MACCROBAT2018 (labels incl. Sign_symptom, Disease_disorder, Medication, Duration, Severity, Dosage)	genuine BioBERT for all 5 categories; great learning	training time; small dataset; case-report domain

13. Alternatives to BioBERT

- **PubMedBERT / BiomedBERT** (Microsoft): pre-trained from scratch on PubMed with its own biomedical vocabulary; often slightly better than BioBERT.
- **ClinicalBERT / Bio_ClinicalBERT**: further trained on MIMIC-III clinical notes - closer to clinical language.
- **scispaCy** (en_ner_bc5cdr_md etc.): fast spaCy pipelines with UMLS linking.
- **MedCAT / cTAKES / MetaMap**: concept extraction with SNOMED/UMLS linking and negation.
- **LLMs**: zero-shot extraction of any category with a prompt, including duration and severity, but with privacy, cost and hallucination concerns; output must be validated against the text.

14. Interview questions and answers

Q1. What is BioBERT?

BERT-base initialised from Google's weights and further pre-trained on PubMed abstracts and PMC articles. Same architecture (12 layers, 768 hidden, 110M parameters) and vocabulary; better biomedical language understanding.

Q2. Can base BioBERT extract symptoms directly?

No. It outputs contextual vectors only. You need a token-classification head fine-tuned on a labelled NER dataset. We used fine-tuned BertForTokenClassification checkpoints.

Q3. Walk me through your extraction pipeline.

Tokenizer -> WordPiece ids and offsets; BioBERT encoder -> 768-d contextual vector per token; linear classifier -> logits over BIO labels; softmax -> probabilities; first-sub-token rule -> word labels; BIO decoding -> entities with character offsets and confidence; attach speaker role.

Q4. What is BIO tagging and why the B tag?

B = beginning, I = inside, O = outside. B separates adjacent entities of the same type, which plain IO tagging cannot.

Q5. How do you handle WordPiece sub-words?

Each word takes the label of its first sub-token (the training convention), and entity spans use the tokenizer's offsets so 'vomit ##ing' maps back to 'vomiting'.

Q6. Which models did you use and why?

alvaroalon2/biobert_diseases_ner and biobert_chemical_ner: genuine fine-tuned BioBERT, documented datasets (BC5CDR, NCBI-disease, BC4CHEMD), Apache-2.0. The most-downloaded BioBERT NER model was for proteins/DNA.

Q7. Why do symptoms come out as DISEASE?

BC5CDR follows MeSH, where signs and symptoms are annotated as disease mentions. So the model finds them but can't distinguish symptom from disease.

Q8. Why didn't you relabel DISEASE as SYMPTOM?

It would misrepresent the model's capability. Mappings must be explicit post-processing steps that can be evaluated, otherwise downstream clinical notes contain unverified claims.

Q9. How would you extract duration and severity?

Short term: transparent rules tagged as rule-based. Properly: fine-tune BioBERT on MACCROBAT2018 which has Duration, Severity, Sign_symptom, Medication labels, and evaluate with seqeval.

Q10. How is the confidence computed and can you trust it?

Average softmax probability of the predicted label across the entity's words. Real model output, but neural networks are over-confident; calibration (e.g. temperature scaling) would be needed to treat it as a probability of correctness.

Q11. Why did the model miss 'my head has been hurting'?

Domain shift: trained on scientific abstracts with terms like 'headache', not lay conversational descriptions. Fine-tuning on conversational clinical data would help.

Q12. Why keep the speaker with each entity?

SOAP needs provenance: patient-reported (Subjective) vs clinician-stated (Assessment/Plan). But speaker isn't ownership - a doctor's question 'any nausea?' answered 'yes' means the patient has nausea.

Q13. What is negation detection and why does it matter?

Deciding whether a mentioned finding is present or absent ('denies chest pain'). Without it, 'no fever' would be recorded as fever. NegEx rules or assertion classifiers are common solutions.

Q14. What is entity linking?

Mapping a mention to a concept ID in an ontology (UMLS, SNOMED CT, RxNorm) so synonyms unify and the data is interoperable. BioSyn and SapBERT are BioBERT-family linkers.

Q15. How would you evaluate this stage?

Annotate a set of consultation transcripts, compute entity-level precision, recall and F1 (strict and partial) per label with seqeval, and analyse errors by type (lay terms, negation, ASR errors).

Q16. BioBERT vs PubMedBERT vs ClinicalBERT?

BioBERT: BERT continued on PubMed/PMC with BERT's vocab. PubMedBERT: from scratch on PubMed with a domain vocab. ClinicalBERT: further trained on clinical notes (MIMIC). Choice depends on whether text is literature-like or clinical-note-like.

Q17. Why run two models instead of one?

No well-documented, licensed BioBERT checkpoint covered both diseases and chemicals; two specialised models are documented and Apache-2.0. Cost: two forward passes and ~860 MB of weights.